Cell Growth: When Less Means More

When is less more? A new study reveals that decreased mitochondrial gene expression and reduced lipid biosynthesis may actually increase cell growth.

Jill Wright and Brandt L. Schneider*

"Being so many different sizes in a day is very confusing," says Alice to the caterpillar, as told by Lewis Carroll in *Alice in Wonderland*. In her simple declaration, Alice eloquently summarizes one of the most fundamental dilemmas in biology. Size matters. Yet, no one size fits all. Remarkably, while organisms display a nearly incomprehensible range of sizes — e.g. the giant Sequoia is nearly 100 billion times larger than the smallest bacterium — cells from similar lineages are strikingly homogeneous in size [1]. As such, organism size is largely dictated by cell number rather than cell size. Humans are larger than mice because we have approximately 3,000 times more cells [2]. Nonetheless, the size of individual cells is often very mutable. A human oocyte can grow to nearly ten times the size of the average cell. How can cell size be both constant and variant? Therein lies the rub. Genetic studies in yeast suggest that the coordination of cell growth with proliferation is essential for cell-size homeostasis [3,4]. However, despite decades of work, the molecular mechanisms that couple growth to cell division remain poorly understood. A number of recent studies have led to the suggestion that a coordinated interplay between cell size and gene expression may be intimately involved in cell size homeostasis [5–9]. Now, elegant results from a new study recently published in *Current Biology* by Miettinen *et al.* [10] enter the fray. Using genomic and metabolomic approaches, the authors have uncovered the somewhat surprising observation that reduced mitochondrial gene expression and decreased lipid synthesis may actually increase cell growth.

The specific molecular mechanisms whereby cell-size homeostasis is maintained have eluded scientists for more than 50 years. On the surface, the solution would appear to be conceptually simple. Balancing cell growth with division rates is sufficient for establishing cell size homeostasis (Figure 1A). However, even minute changes will rapidly disrupt the balance. Decreasing growth rate without a compensatory reduction in proliferation rates will decrease cell size (Figure 1B). The converse is also true: increasing growth rate *in lieu* of increased cell division rates will produce large cells (Figure 1C). Therefore, cells must have a means for balancing growth with division. The yeast paradigm proposes that cell-cycle progression is blocked until cells attain a minimum size [11]. In this manner, proliferation is coupled to cell size. While genetic studies have identified a number of highly conserved genes that are integrally involved in linking cell size to proliferative capacity, a number of questions still remain [12,13]. How does a cell know how big it is? How does a cell know how big it should be? How does growth to a specific cell size trigger cell-cycle progression?

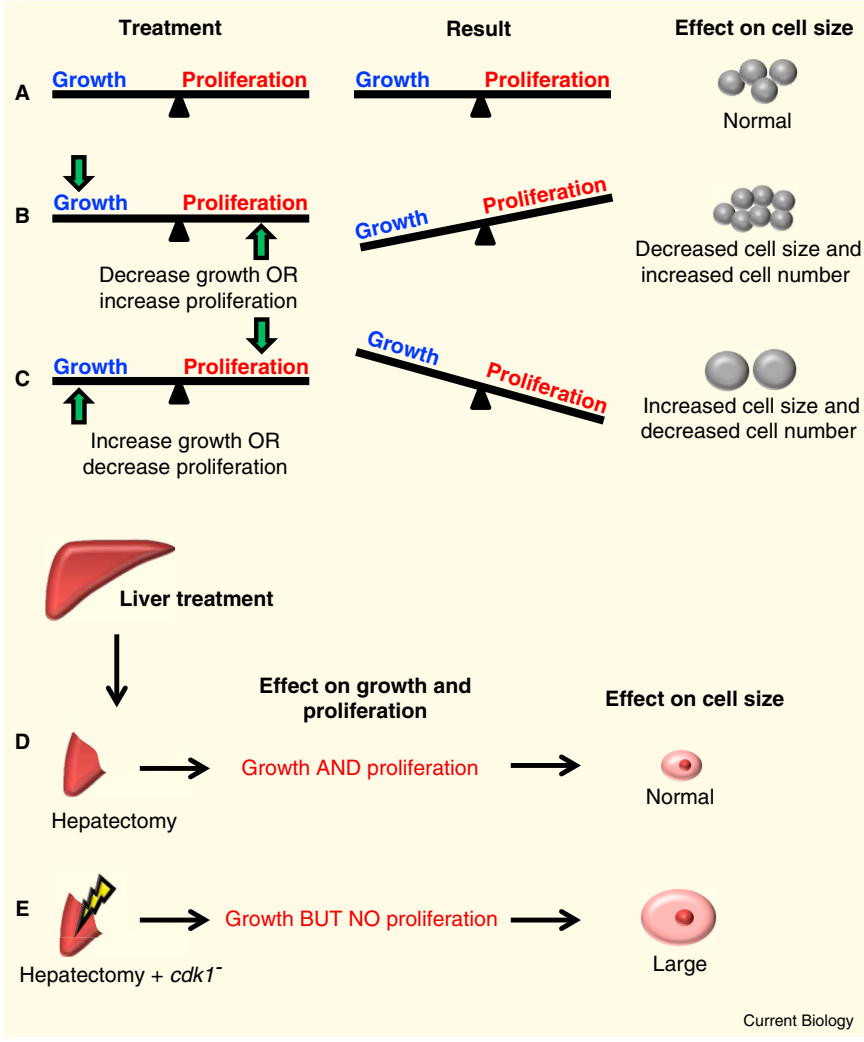


Figure 1. Effects of growth and proliferation.

(A) Cell-size homeostasis is dependent upon a balance between growth and proliferation. (B) A decrease in growth or an increase in proliferation results in a reduction of cell size. (C) In contrast, an increase in growth or a decrease in proliferation produces large cells. (D,E) In normal untreated liver, hepatocytes are quiescent — i.e., they do not grow or divide. (D) Following a partial hepatectomy, hepatocytes are stimulated to grow and divide in a balanced manner, so cell size remains unchanged. (E) However, when proliferation is blocked (e.g. in cells lacking Cdk1), organ regeneration after a partial hepatectomy occurs predominately by cell growth and results in severe hypertrophy.

CrossMark

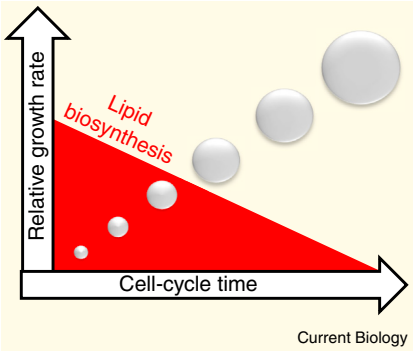


Figure 2. Cell size inversely correlates with lipid levels.

Reduction of mitochondrial metabolism and lipid synthesis (shown by the red triangle) results in increased cell-cycle time (i.e. decreased proliferation rates) and increased relative cell growth, producing abnormally large cells.

Miettinen *et al.* [10] used a clever approach to address these questions. The liver is a remarkably regenerative organ. Removal of nearly three-quarters of a mouse liver — i.e. partial hepatectomy — induces the remaining hepatocytes to grow and proliferate to replace the lost tissue (Figure 1D). Under these conditions, cell-size homeostasis is maintained. Miettinen *et al.* [10] ablated expression of cyclin-dependent kinase 1 (Cdk1) specifically in adult mouse liver hepatocytes; since Cdk1 is essential for cell-cycle progression, hepatocytes lacking Cdk1 are unable to proliferate. Nonetheless, these authors found that blocking proliferation had little impact on cell growth: the end result was that excessive growth (hypertrophy) produced abnormally large cells that fully regenerated the liver (Figure 1E). After utilizing this approach to disconnect cell size from proliferative capacity, the authors subsequently used RNA sequencing to examine how cell size differentially affects gene expression. Very few studies have examined the relationship between cell size and gene expression in metazoans [10,14]. As expected, larger cells upregulated structural genes involved in cell growth, but unpredictably, many genes involved in mitochondrial function were significantly repressed in these cells [10]. This was particularly surprising, given that previous studies have indicated that the loss of mitochondrial genes is closely associated with decreased rather than increased cell size [12,13]. Importantly, control experiments confirmed that the repression of mitochondrial genes was a size-dependent effect. In addition, complementary experiments performed in *Drosophila* Kc167 cells substantiated the observation that cellular hypertrophy represses mitochondrial gene expression [10].

The authors backed up this impressive array of gene expression data with a systematic metabolomics approach. Repression of mitochondrial gene expression suggested that mitochondrial structure or function might be impacted. Interestingly, larger cells displayed no obvious mitochondrial defects or decrease in ATP production [10]. In contrast, metabolic data indicated that increased cell size concomitantly upregulated glycolysis. However, what sets this new work apart is the examination of the impact of pharmacological mitochondrial inhibitors on the size and proliferative capacity of cells.

By using a panel of small-molecule inhibitors that target mitochondria and/or repress metabolic pathways involved in glycolysis and the pentose phosphate pathway, Miettinen *et al.* [10] progress from descriptive observations to a mechanistic approach. Strikingly, the authors found that inhibiting mitochondrial function (e.g., the use of minocycline or thiostrepton to inhibit mitochondrial translation, Mdivi-1 to inhibit mitochondrial fission, or the uncoupling agents FCCP and CCCP) increased cell size and decreased proliferative capacity. Additional studies revealed that cultured cells that lacked mitochondrial DNA were also larger than normal. However, inhibitors that blocked mitochondrial pathways involved in oxidative phosphorylation did not result in cell-size increases [10]. Therefore, the authors investigated whether additional functions linked mitochondrial metabolism to cell-size control.

Another key mitochondrial function is the production of acetyl-coenzyme A, a precursor for lipid biosynthesis. Metabolomics indicated that key mitochondrial transporters (e.g., the citrate transporter SLC25A1 and the pyruvate transporter BRP44) were repressed in large cells. In addition, depletion of SLC25A1 and its transcriptional activator PGC-1a recapitulated the size results [10]. However, providing these cells with a cocktail of commercially produced lipids rescued the size defect. Moreover, the addition of lipids also promoted proliferation and reduced size in untreated cells [10], in line with other studies that have demonstrated a lipid-synthesis requirement for cell-cycle progression [15]. Additional investigations by Miettinen *et al.* [10] revealed that several key transcription factors involved in lipid biosynthesis were downregulated in large cells. Furthermore, pharmacological inhibition of lipid biosynthesis increased cell size and decreased proliferation rates. The take-home message is that mitochondrial function and lipid biosynthesis are integrally involved in balancing cell growth with proliferation, leading to the proposal of an intriguing model (Figure 2). In this hypothetical feedback inhibition model, excess lipids repress fatty acid biosynthesis, which results in reduced mitochondrial function, decreased proliferative potential, and increased cell size. Implicit in this model is the possibility that cells use the build-up of unused lipids to sense cell size. However, one problem with such a model is that it is difficult to determine whether lipid levels are modulating cell size or *vice versa*.

The very interesting yet unexpected observations made by Miettinen *et al.* [10] provide new fodder for further studies to address a number of intriguing questions. First, yeast studies suggest that a decrease in mitochondrial function reduces rather than increases cell size [12,13]: how can this difference be explained? Second, it is widely believed that a decline in transport efficiency and diffusion-limited processes result in energy deprivation, which in turn limits cell size [1]. There is no evidence from these studies that the recovering hepatocytes are bioenergetically challenged. Therefore, it appears that physics of cell growth may directly impact gene expression and metabolic regulation. However, these effects are remarkably complex and may involve the molecular sensing of specific lipid species. What molecules are involved, and would they stimulate proliferation in physiologically quiescent cells? Finally, there is growing evidence that hypertrophy can limit the lifespan of cells [16–19]. Thus, when organ recovery is dominated by increases in cell size rather than cell number is there a detrimental trade-off? The hypothesis

that the availability of lipids may regulate proliferative potential is also entering the clinical arena where ongoing studies are evaluating the efficacy of lipid-lowering statins as anti-cancer drugs [14,20]. These intriguing new observations by Miettinen *et al.* [10] reveal that the molecular and genetic pathways that balance cell growth with proliferation are perhaps even more complicated than previously suspected.

References

1. Bonner, J.T. (2006). Why Size Matters: From Bacteria to Blue Whales (Princeton: Princeton University Press).
2. Conlon, I., and Raff, M. (1999). Size control in animal development. Cell *96*, 235–244.
3. Turner, J.J., Ewald, J.C., and Skotheim, J.M. (2012). Cell size control in yeast. Curr. Biol. *22*, R350–R359.
4. Jorgensen, P., and Tyers, M. (2004). How cells coordinate growth and division. Curr. Biol. *14*, R1014–R1027.
5. Marguerat, S., and Bahler, J. (2012). Coordinating genome expression with cell size. Trends Genet. *28*, 560–565.
6. Wu, C.Y., Rolfe, P.A., Gifford, D.K., and Fink, G.R. (2010). Control of transcription by cell size. PLoS Biol. *8*, e1000523.
7. Dungrawala, H., Manukyan, A., and Schneider, B.L. (2010). Gene regulation: global transcription rates scale with size. Curr. Biol. *20*, R979–R981.
8. Bonke, M., Turunen, M., Sokolova, M., Vähärautio, A., Kivioja, T., Taipale, M., Björklund, M., and Taipale, J. (2013). Transcriptional networks controlling the cell cycle. G3 *3*, 75–90.
9. Zhurinsky, J., Leonhard, K., Watt, S., Marguerat, S., Bähler, J., and Nurse, P. (2010). A coordinated global control over cellular transcription. Curr. Biol. *20*, 2010–2015.
10. Miettinen, T.P., Pessa, H.K.J., Caldez, M.J., Fuhrer, T., Diril, M.K., Sauer, U., Kaldis, P., and Björklund, M. (2014). Identification of transcriptional and metabolic programs related to mammalian cell size. Curr. Biol. *24*, 598–608.
11. Hartwell, L.H. (1974). Saccharomyces cerevisiae cell cycle. Bacteriol. Rev. *38*, 164–198.
12. Zhang, J., Schneider, C., Ottmers, L., Rodriguez, R., Day, A., Markwardt, J., and Schneider, B.L. (2002). Genomic scale mutant hunt identifies cell size homeostasis genes in S. cerevisiae. Curr. Biol. *12*, 1992–2001.
13. Jorgensen, P., Nishikawa, J.L., Breitkreutz, B.J., and Tyers, M. (2002). Systematic identification of pathways that couple cell growth and division in yeast. Science *297*, 395–400.
14. Dolfi, S.C., Chan, L.L., Qiu, J., Tedeschi, P.M., Bertino, J.R., Hirshfield, K.M., Oltvai, Z.N., and Vazquez, A. (2013). The metabolic demands of cancer cells are coupled to their size and protein synthesis rates. Cancer Metab. *1*, 20.
15. Kwok, A.C., and Wong, J.T. (2005). Lipid biosynthesis and its coordination with cell cycle progression. Plant Cell Physiol. *46*, 1973–1986.
16. Yang, J., Dungrawala, H., Hua, H., Manukyan, A., Abraham, L., Lane, W., Mead, H., Wright, J., and Schneider, B.L. (2011). Cell size and growth rate are major determinants of replicative lifespan. Cell Cycle *10*, 144–155.
17. Gems, D., and Partridge, L. (2013). Genetics of longevity in model organisms: debates and paradigm shifts. Annu. Rev. Physiol. *75*, 621–644.
18. Wright, J., Dungrawala, H., Bright, R.K., and Schneider, B.L. (2013). A growing role for hypertrophy in senescence. FEMS Yeast Res. *13*, 2–6.
19. Blagosklonny, M.V. (2012). Answering the ultimate question "what is the proximal cause of aging?" Aging *4*, 861–877.
20. Nielsen, S.F., Nordestgaard, B.G., and Bojesen, S.E. (2012). Statin use and reduced cancer-related mortality. N. Engl. J. Med. *367*, 1792–1802.

3601 4th St Rm 5C119, Department of Cell Biology and Biochemistry, Texas Tech University, Health Sciences Center, Lubbock, TX 79430, USA.
*E-mail: brandt.schneider@ttuhsc.edu

http://dx.doi.org/10.1016/j.cub.2014.02.044

Neurobiology: Sensory Lateralization in the Fish Brain

In zebrafish, the dorsal habenula shows conspicuous left-right differences. New research shows that the left and right habenula differentially process visual and olfactory information. Spontaneous activity in habenular circuits may lead to activation of distinct neuronal targets and behavioral programs.

Hitoshi Okamoto

The majority of the human population can control the right hand more skillfully than the left hand. Likewise, the left half of the human face is in most cases better at expressing a smile than the right half [1]. From such observations, it has been inferred that the left and right hemispheres of the human brain show functional differences. In the 19th century, Dax and Broca discovered a lesion in the left hemisphere of post-mortem brains in patients with severe impairment in speech ability [2,3], and subsequently, Wernicke found that a damage to another region of the left hemisphere was associated with an impairment of language comprehension [4]. These discoveries for the first time provided the evidence that human language ability is controlled by the two distinct areas of the left cortex. In the 20th century, by careful observation of the patients with a split brain in which the corpus callosum, the nerve fiber bundles connecting the left and right hemispheres reciprocally, was surgically severed, Sperry demonstrated that — by and large — the left hemisphere is specialized for logical thinking and language processing, while the right hemisphere is more adapted for shape recognition, or emotional and artistic functions [5]. It certainly might be a more efficient and universal strategy for expanding the capacity of the brain to have two hemispheres of the brain engaged in information processing of different categories rather than having them dedicated to redundantly same subjects. Indeed, functional asymmetry of the brain has been found in other animals, too. For example, the two hemispheres of birds display a complementary pattern of visual analysis. The left hemisphere is specialized for detailed object analysis, attends to local features and excels in the categorization of visual stimuli. In contrast, the right hemisphere extracts relational configurations of visual stimuli that can be relevant during spatial orientation [6]. Now, two reports by Dreosti *et al.* [7] and Jetti *et al.* [8] in this issue of *Current Biology* demonstrate that the left and right habenula in zebrafish have a distinct difference in the modality of sensory information which they process, *i.e.* the left and right dHb differentially process visual and olfactory information, respectively.

Although some areas of the human brain, such as the planum temporale [9], show anatomical asymmetry and several molecules such as LMO4 have been identified to be expressed asymmetrically in both hemispheres [10], it has been difficult to decisively attribute genes to the establishment of brain asymmetry only by studying humans. It is evident that genetically tractable model animals are required to understand the functional, anatomical and genetic basis of brain laterality. However, this has been difficult in mice,

CrossMark